定义（$\mathbb{R}^2$上的刚体运动）$\mathbb{R}^2$是$\mathbb{R}$-向量空间，

对$\forall x=\begin{pmatrix}x_1\\x_2\end{pmatrix}\in\mathbb{R}^2$，$\forall y=\begin{pmatrix}y_1\\y_2\end{pmatrix}\in\mathbb{R}^2$，定义$x$和$y$的标准内积

为$(x|y)=x_1y_1+x_2y_2$

对$\forall x\in\mathbb{R}^2$，定义x的Euclid长度为$\|x\|=\sqrt{(x|x)}\in\mathbb{R}_{\geq 0}$

$f:\mathbb{R}^2\to\mathbb{R}^2$是一个任意的映射．若对$\forall x,y\in\mathbb{R}^2$，有：

$\|f(x)-f(y)\|=\|x-y\|$，则称 映射$f:\mathbb{R}^2\to\mathbb{R}^2$是$\mathbb{R}^2$上的刚体运动．

Lemma：$\mathbb{R}^2$是$\mathbb{R}$-向量空间，$f:\mathbb{R}^2\to\mathbb{R}^2$是一个任意的映射．则下列命题等价：

① 映射$f:\mathbb{R}^2\to\mathbb{R}^2$是$\mathbb{R}^2$上的刚体运动．

② 存在唯一的2×2实正交矩阵A和唯一的$b\in\mathbb{R}^2$，使得对$\forall x\in\mathbb{R}^2$，都有$f(x)=Ax+b$．

Proof：上篇笔记已证．$\square$

定义（$\mathbb{R}^2$上的平移）$\mathbb{R}^2$是$\mathbb{R}$-向量空间，$f:\mathbb{R}^2\to\mathbb{R}^2$是一个任意的映射．若存在$b\in\mathbb{R}^2$，使得对$\forall x\in\mathbb{R}^2$，都有$f(x)=x+b$，则称映射$f:\mathbb{R}^2\to\mathbb{R}^2$是$\mathbb{R}^2$上的平移．

Lemma：$\mathbb{R}^2$上的平移是$\mathbb{R}^2$上的刚体运动.

Proof：设映射$f: \mathbb{R}^2 \to \mathbb{R}^2$是$\mathbb{R}^2$上的平移. 则存在$b \in \mathbb{R}^2$，使得对$\forall x \in \mathbb{R}^2$，都有$f(x) = x + b$

$\therefore$ 对$\forall x, y \in \mathbb{R}^2$，有：$\|f(x) - f(y)\| = \|(x+b) - (y+b)\|$

$= \|x + b - y - b\| = \|x - y\|$

$\therefore$ 映射$f: \mathbb{R}^2 \to \mathbb{R}^2$是$\mathbb{R}^2$上的刚体运动. $\square$

Lemma：$\mathbb{R}^2$是$\mathbb{R}$一向量空间，$f: \mathbb{R}^2 \to \mathbb{R}^2$是一个任意的映射，存在$A \in M_{2\times 2}(\mathbb{R})$和$b \in \mathbb{R}^2$，使得对$\forall x \in \mathbb{R}^2$，都有$f(x) = Ax + b$.

则下列命题等价：

① 映射$f: \mathbb{R}^2 \to \mathbb{R}^2$是$\mathbb{R}^2$上的平移

② $A = I_{2\times 2}$

Proof：②$\Rightarrow$①：已知存在$A \in M_{2\times 2}(\mathbb{R})$和$b \in \mathbb{R}^2$，s.t. 对$\forall x \in \mathbb{R}^2$，都有$f(x) = Ax + b$.

$\because A = I_{2\times 2}$ $\therefore$ 对$\forall x \in \mathbb{R}^2$，都有$f(x) = I_{2\times 2}x + b$

$\therefore$ 对$\forall x \in \mathbb{R}^2$，都有$f(x) = x + b$

$\therefore$ 映射$f: \mathbb{R}^2 \to \mathbb{R}^2$是$\mathbb{R}^2$上的平移.

①$\Rightarrow$②：$\because$ 映射$f: \mathbb{R}^2 \to \mathbb{R}^2$是$\mathbb{R}^2$上的平移

$\therefore$ 存在$t \in \mathbb{R}^2$，s.t. 对$\forall x \in \mathbb{R}^2$，都有$f(x) = x + t$.

$\because$ 存在 $A \in M_{2\times 2}(\mathbb{R})$ 和 $b \in \mathbb{R}^2$, s.t. 对 $\forall x \in \mathbb{R}^2$, 都有 $f(x) = Ax + b$

$\therefore$ 对 $\forall x \in \mathbb{R}^2$, 都有 $f(x) = x + t$ 且 $f(x) = Ax + b$

$\therefore$ 对 $\forall x \in \mathbb{R}^2$, 都有 $x + t = Ax + b$

$\because \begin{pmatrix} 0 \\ 0 \end{pmatrix} \in \mathbb{R}^2 \quad \therefore \begin{pmatrix} 0 \\ 0 \end{pmatrix} + t = A \cdot \begin{pmatrix} 0 \\ 0 \end{pmatrix} + b \qquad \therefore t = b$

$\therefore$ 对 $\forall x \in \mathbb{R}^2$, 都有 $x + t = Ax + t$

$\therefore$ 对 $\forall x \in \mathbb{R}^2$, 都有 $x = Ax$.

设 $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \qquad \therefore$ 对 $\forall x \in \mathbb{R}^2$, 都有 $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} x = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} x$

$\because \begin{pmatrix} 1 \\ 0 \end{pmatrix} \in \mathbb{R}^2 \quad \therefore \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \therefore \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix}$

$\therefore a_{11} = 1$ 且 $a_{21} = 0$

$\because \begin{pmatrix} 0 \\ 1 \end{pmatrix} \in \mathbb{R}^2 \quad \therefore \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \qquad \therefore \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix}$

$\therefore a_{12} = 0$ 且 $a_{22} = 1$

$\therefore A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_{2\times 2} \qquad \square$

Lemma: $\mathbb{R}^2$ 是 $\mathbb{R}$-向量空间，映射 $f: \mathbb{R}^2 \to \mathbb{R}^2$ 是 $\mathbb{R}^2$ 上的刚体运动，已经证明了存在唯一的 2×2 实正交矩阵 A 和唯一的 $b \in \mathbb{R}^2$，使得对 $\forall x \in \mathbb{R}^2$，都有 $f(x) = Ax + b$. 则下列命题等价:

① 映射 $f: \mathbb{R}^2 \to \mathbb{R}^2$ 是 $\mathbb{R}^2$ 上的平移

② $A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ 且 $\theta = 2k\pi$, $k \in \mathbb{Z}$.

Proof: $\because f:\mathbb{R}^2\to\mathbb{R}^2$ 是映射，$A\in M_{2\times2}(\mathbb{R})$，$b\in\mathbb{R}^2$，

对 $\forall x\in\mathbb{R}^2$ 都有 $f(x)=Ax+b$

$\therefore$ ① $\Longleftrightarrow A=I_{2\times2}$，已知 A 是 2×2 实正交矩阵

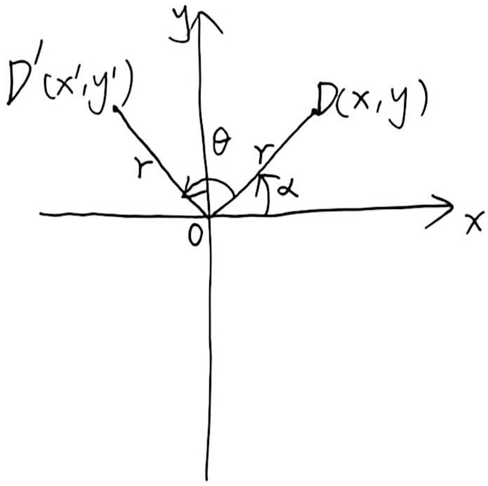
假设 $A=\begin{pmatrix}\cos\theta & \sin\theta\\ \sin\theta & -\cos\theta\end{pmatrix}$，则 $\because A=I_{2\times2}$ $\therefore \begin{pmatrix}\cos\theta & \sin\theta\\ \sin\theta & -\cos\theta\end{pmatrix}=\begin{pmatrix}1&0\\0&1\end{pmatrix}$

$\therefore \cos\theta=1$ 且 $-\cos\theta=1$ $\quad\therefore 1=-1$ 矛盾. $\quad\therefore A\neq\begin{pmatrix}\cos\theta & \sin\theta\\ \sin\theta & -\cos\theta\end{pmatrix}$

$\therefore$ ① $\Longleftrightarrow A=I_{2\times2}\Longleftrightarrow A=\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$ 且 $\cos\theta=1$ 且 $\sin\theta=0$

$\Longleftrightarrow A=\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$ 且 $\theta=2k\pi$，$k\in\mathbb{Z}$. $\square$

例：平面直角坐标系中任意一点 (x,y) 绕原点逆时针旋转 $\theta\in\mathbb{R}$ 后的点的坐标为 (x',y'). 求 (x',y').



解：设点 (x,y) 与原点 O 的距离为 r. $\because$ 点 (x,y) 绕原点 O 逆时针旋转 θ 到点 (x',y') $\quad\therefore$ 点 (x',y') 与原点 O 的距离为 r.

设 $D(x,y)$，$D'(x',y')$. 设 $\overrightarrow{OD}$ 与 x 轴正向所夹的有向角为 α. $(\alpha\in\mathbb{R})$

$\therefore \overrightarrow{OD'}$ 与 x 轴正向所夹的有向角为 $\alpha+\theta$

$\because D(x,y) \quad \therefore \frac{x}{r}=\cos\alpha,\ \frac{y}{r}=\sin\alpha$

$\therefore x=r\cdot\cos\alpha,\ y=r\cdot\sin\alpha$

$\because D'(x',y') \quad \therefore \frac{x'}{r}=\cos(\alpha+\theta),\ \frac{y'}{r}=\sin(\alpha+\theta)$

$\therefore x'=r\cdot\cos(\alpha+\theta),\ y'=r\cdot\sin(\alpha+\theta)$

$$\begin{aligned}\because x'=r\cdot\cos(\alpha+\theta) &= r(\cos\alpha\cos\theta-\sin\alpha\sin\theta)\\ &= r\cos\alpha\cos\theta - r\sin\alpha\sin\theta = x\cdot\cos\theta - y\cdot\sin\theta\\ &= \cos\theta\cdot x + (-\sin\theta)\cdot y\end{aligned}$$

$$\begin{aligned}y'=r\cdot\sin(\alpha+\theta) &= r\cdot(\sin\alpha\cos\theta+\cos\alpha\sin\theta)\\ &= r\sin\alpha\cos\theta + r\cos\alpha\sin\theta = y\cdot\cos\theta + x\cdot\sin\theta\\ &= \sin\theta\cdot x + \cos\theta\cdot y\end{aligned}$$

$$\therefore \begin{pmatrix}x'\\y'\end{pmatrix}=\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} \qquad \square$$

定义（旋转矩阵）对$\forall\theta\in\mathbb{R}$，定义实$2\times2$矩阵

$$R(\theta):=\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$$

称矩阵$R(\theta)$为2×2旋转矩阵.

Lemma：对$\forall\theta_1,\theta_2\in\mathbb{R}$，有：$R(\theta_1+\theta_2)=R(\theta_1)R(\theta_2)=R(\theta_2)R(\theta_1)$

Proof：
$$R(\theta_1)R(\theta_2)=\begin{pmatrix}\cos\theta_1 & -\sin\theta_1\\ \sin\theta_1 & \cos\theta_1\end{pmatrix}\begin{pmatrix}\cos\theta_2 & -\sin\theta_2\\ \sin\theta_2 & \cos\theta_2\end{pmatrix}$$

$$=\begin{pmatrix}\cos\theta_1\cos\theta_2-\sin\theta_1\sin\theta_2 & -\cos\theta_1\sin\theta_2-\sin\theta_1\cos\theta_2\\ \sin\theta_1\cos\theta_2+\cos\theta_1\sin\theta_2 & \cos\theta_1\cos\theta_2-\sin\theta_1\sin\theta_2\end{pmatrix}$$

$$= \begin{pmatrix} \cos(\theta_1+\theta_2) & -\sin(\theta_1+\theta_2) \\ \sin(\theta_1+\theta_2) & \cos(\theta_1+\theta_2) \end{pmatrix} = R(\theta_1+\theta_2)$$

$$R(\theta_2)R(\theta_1) = \begin{pmatrix} \cos\theta_2 & -\sin\theta_2 \\ \sin\theta_2 & \cos\theta_2 \end{pmatrix}\begin{pmatrix} \cos\theta_1 & -\sin\theta_1 \\ \sin\theta_1 & \cos\theta_1 \end{pmatrix}$$

$$= \begin{pmatrix} \cos\theta_2\cos\theta_1 - \sin\theta_2\sin\theta_1 & -\sin\theta_1\cos\theta_2 - \cos\theta_1\sin\theta_2 \\ \sin\theta_2\cos\theta_1 + \cos\theta_2\sin\theta_1 & \cos\theta_2\cos\theta_1 - \sin\theta_2\sin\theta_1 \end{pmatrix}$$

$$= \begin{pmatrix} \cos(\theta_1+\theta_2) & -\sin(\theta_1+\theta_2) \\ \sin(\theta_1+\theta_2) & \cos(\theta_1+\theta_2) \end{pmatrix} = R(\theta_1+\theta_2)$$

$$\therefore R(\theta_1)R(\theta_2) = R(\theta_1+\theta_2) = R(\theta_2)R(\theta_1) \qquad \square$$

Lemma：对$\forall \theta \in \mathbb{R}$，有：$R(\theta)$是$2\times 2$实正交矩阵，且$\det(R(\theta)) = 1$，且$(R(\theta))^{-1} = R(-\theta)$

Proof：对$\forall \theta \in \mathbb{R}$，$\because R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ $\therefore R(\theta)$是2×2实正交矩阵.

$$\det(R(\theta)) = \cos^2\theta + \sin^2\theta = 1$$

$\because R(\theta)$是2×2实正交矩阵

$$\therefore (R(\theta))^{-1} = {}^t(R(\theta)) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{pmatrix} = R(-\theta)$$

$\square$

Lemma：对$\forall \theta \in \mathbb{R}$，有：

① $R(\theta) = I_{2\times 2} \iff \theta = 2k\pi,\ k \in \mathbb{Z}$

② $R(\theta) = -I_{2\times 2} \iff \theta = \pi + 2k\pi,\ k \in \mathbb{Z}$

Proof: ①: $R(\theta)=I_{2\times 2} \iff \begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix}=\begin{pmatrix}1 & 0 \\ 0 & 1\end{pmatrix}$

$\iff \cos\theta=1$ 且 $\sin\theta=0 \iff \theta=2k\pi,\ k\in\mathbb{Z}$

②: $R(\theta)=-I_{2\times 2} \iff \begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix}=\begin{pmatrix}-1 & 0 \\ 0 & -1\end{pmatrix}$

$\iff \cos\theta=-1$ 且 $\sin\theta=0 \iff \theta=\pi+2k\pi,\ k\in\mathbb{Z}$ $\square$

~~Lemma~~ 定义：对 $\forall\, \theta_1, \theta_2 \in \mathbb{R}$，若 $\theta_1-\theta_2 \in 2\pi\mathbb{Z}$，则记

$\theta_1 \equiv \theta_2 \pmod{2\pi}$.

显然，这是 $\mathbb{R}$ 上的等价关系. 含 θ 的等价类也记为 $\theta \bmod 2\pi$.

Lemma: 对 $\forall\, \theta_1, \theta_2 \in \mathbb{R}$，有：

$R(\theta_1)=R(\theta_2) \iff e^{i\theta_1}=e^{i\theta_2} \iff \theta_1 \equiv \theta_2 \pmod{2\pi}$

Proof: $R(\theta_1)=R(\theta_2) \iff \begin{pmatrix}\cos\theta_1 & -\sin\theta_1 \\ \sin\theta_1 & \cos\theta_1\end{pmatrix}=\begin{pmatrix}\cos\theta_2 & -\sin\theta_2 \\ \sin\theta_2 & \cos\theta_2\end{pmatrix}$

$\iff \cos\theta_1=\cos\theta_2$ 且 $\sin\theta_1=\sin\theta_2$

$\iff \cos\theta_1+i\sin\theta_1=\cos\theta_2+i\sin\theta_2 \iff e^{i\theta_1}=e^{i\theta_2}$

$\iff$ 角 θ_1 与角 θ_2 的终边相同 $\iff \theta_1-\theta_2 \in 2\pi\mathbb{Z}$

$\iff \theta_1 \equiv \theta_2 \pmod{2\pi}$ $\square$

Lemma: P 是 2×2 实正交矩阵. 记 $\varepsilon=\det(P)$. 则有：

对 $\forall\, \theta\in\mathbb{R}$，$P^{-1}R(\theta)P=R(\varepsilon\theta)$

Proof: ~~分两种情况讨论~~ $\because P$ 是 2×2 实正交矩阵 $\therefore$ 分2种情况讨论.

① $P=\begin{pmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{pmatrix}$ (其中 $\alpha\in\mathbb{R}$). $\therefore \det(P)=1=\varepsilon$

$\therefore P=R(\alpha)$. $\quad P^{-1}=(R(\alpha))^{-1}=R(-\alpha)$

$\therefore P^{-1}R(\theta)P=R(-\alpha)R(\theta)R(\alpha)=R(-\alpha+\theta+\alpha)$

$=R(\theta)=R(1\cdot\theta)=R(\varepsilon\theta)$

② $P=\begin{pmatrix}\cos\alpha & \sin\alpha\\ \sin\alpha & -\cos\alpha\end{pmatrix}$ (其中 $\alpha\in\mathbb{R}$) $\therefore \det(P)=-1=\varepsilon$

$\therefore P\cdot\begin{pmatrix}1&0\\0&-1\end{pmatrix}=\begin{pmatrix}\cos\alpha & \sin\alpha\\ \sin\alpha & -\cos\alpha\end{pmatrix}\begin{pmatrix}1&0\\0&-1\end{pmatrix}=\begin{pmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{pmatrix}=R(\alpha)$

$\because \begin{pmatrix}1&0\\0&-1\end{pmatrix}\begin{pmatrix}1&0\\0&-1\end{pmatrix}=\begin{pmatrix}1&0\\0&1\end{pmatrix}=I_{2\times2}$

$\therefore P=R(\alpha)\cdot\begin{pmatrix}1&0\\0&-1\end{pmatrix}$ $\quad P^{-1}=\begin{pmatrix}1&0\\0&-1\end{pmatrix}R(-\alpha)$

$\therefore P^{-1}R(\theta)P=\begin{pmatrix}1&0\\0&-1\end{pmatrix}R(-\alpha)R(\theta)R(\alpha)\begin{pmatrix}1&0\\0&-1\end{pmatrix}$

$=\begin{pmatrix}1&0\\0&-1\end{pmatrix}R(-\alpha+\theta+\alpha)\begin{pmatrix}1&0\\0&-1\end{pmatrix}=\begin{pmatrix}1&0\\0&-1\end{pmatrix}R(\theta)\begin{pmatrix}1&0\\0&-1\end{pmatrix}$

$=\begin{pmatrix}1&0\\0&-1\end{pmatrix}\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}1&0\\0&-1\end{pmatrix}=\begin{pmatrix}\cos\theta & -\sin\theta\\ -\sin\theta & -\cos\theta\end{pmatrix}\begin{pmatrix}1&0\\0&-1\end{pmatrix}$

$=\begin{pmatrix}\cos\theta & \sin\theta\\ -\sin\theta & \cos\theta\end{pmatrix}=\begin{pmatrix}\cos(-\theta) & -\sin(-\theta)\\ \sin(-\theta) & \cos(-\theta)\end{pmatrix}=R(-\theta)=R(\varepsilon\theta)$ $\square$